
Problem 1. *Use* ***Cylindrical coordinates*** *and Gauss's Divergence Theorem to compute the flux integral* $\iint \vec{F} \cdot \vec{n}\, dA$ *for the vector field (in Cartesian coordinates)* $\vec{F} = [xy, yz, zx]$*, where* S *is, in Cartesian coordinates, the surface of the cone* $x^2 + y^2 \leq 4z$, $0 \leq z \leq 2$.

Note: In this problem, you are expected to:

- *write down Gauss's Divergence Theorem,*
- *set up an appropriate triple integral using the Cartesian coordinates,*
- *write down the transformation functions between Cartesian and Cylindrical coordinates,*
- *recall the Jacobian* $\frac{\partial(x,y,z)}{\partial(r,\theta,z)}$ *required for the change of variables in the integral, and*
- *evaluate the final result.*

Problem 2. *For the* ***Cylindrical coordinates****, compute both the Derivative Matrices for transforming from Cartesian to Cylindrical and vice-versa.*

Then compute the Jacobian $\frac{\partial(x,y,z)}{\partial(r,\theta,z)}$.

Finally, compute the appropriate Lamé coefficients and use them to write the expression for $\nabla \cdot \vec{F}$ *in terms of* r, θ, z*, where*

$$\vec{F} = F_r \hat{e}_r + F_\theta \hat{e}_\theta + F_z \hat{e}_z.$$

You may use the fact:

$$\nabla \cdot \vec{F} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial r}(h_2 h_3 F_r) + \frac{\partial}{\partial \theta}(h_3 h_1 F_\theta) + \frac{\partial}{\partial z}(h_1 h_2 F_z) \right].$$

Problem 3. *For the* ***Spherical coordinates*** *compute the appropriate Lamé coefficients and use them to write the expression for* $\nabla \times \vec{F}$ *in terms of* ρ, θ, ϕ*, where*

$$\vec{F} = F_\rho \hat{e}_\rho + F_\theta \hat{e}_\theta + F_\phi \hat{e}_\phi.$$

You may use the fact:

$$\nabla \times \vec{F} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{e}_\rho & h_2 \hat{e}_\theta & h_3 \hat{e}_\phi \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ h_1 F_\rho & h_2 F_\theta & h_3 F_\phi \end{vmatrix}.$$

Problem 4. *Use the appropriate Lamé coefficients to compute the Laplacian of the scalar function* $f(x, y, z) = x^2 z + y^2 z + z^3$ *in both Cylindrical and Spherical coordinates.*

Recall: for generic coordinates (q_1, q_2, q_3)*:*

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial q_1} \right) + \frac{\partial}{\partial q_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial f}{\partial q_2} \right) + \frac{\partial}{\partial q_1} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial q_3} \right) \right].$$